NAME

rgbasm — Game Boy assembler

SYNOPSIS

rgbasm [−EhVw] [−v [−v ...]] [−B *param*] [−b *chars*] [−−color *when*]
[−D *name*[=*value*]] [−g *chars*] [−I *path*] [−M *depend_file*] [−MG] [−MC] [−MP]
[−MT *target_file*] [−MQ *target_file*] [−o *out_file*] [−P *include_file*]
[−p *pad_value*] [−Q *fix_precision*] [−r *recursion_depth*]
[−s *features*:*state_file*] [−W *warning*] [−X *max_errors*] *asmfile*

DESCRIPTION

The **rgbasm** program creates an RGB object file from an assembly source file. The object file format is documented in *rgbds*(5).

ARGUMENTS

rgbasm accepts the usual short and long options, such as `-V` and `--version`. Options later in the command line override those set earlier, except for when duplicate options are considered an error. Options can be abbreviated as long as the abbreviation is unambiguous: `--verb` is `--verbose`, but `--ver` is invalid because it could also be `--version`.

Unless otherwise noted, passing '`-`' (a single dash) as a file name makes **rgbasm** use standard input (for input files) or standard output (for output files). To suppress this behavior, and open a file in the current directory actually called '`-`', pass `./-` instead. Using standard input or output for more than one file in a single command may produce unexpected results.

rgbasm accepts decimal, hexadecimal, octal, and binary for numeric option arguments. Decimal numbers are written as usual; hexadecimal numbers must be prefixed with either '`$`' or '`0x`'; octal numbers must be prefixed with either '`&`' or '`0o`'; and binary numbers must be prefixed with either '`%`' or '`0b`'. (The prefixes '`$`' and '`&`' will likely need escaping or quoting to avoid being interpreted by the shell.) Leading zeros (after the base prefix, if any) are accepted, and letters are not case-sensitive. For example, all of these are equivalent: '`42`', `042`, `0x2A`, `0X2A`, `0x2a`, `&52`, `0o52`, `0O052`, `0b00101010`, `0B101010`.

The following options are accepted:

`-B` *param*, `--backtrace` *param*
: Configures how location backtraces are printed if warnings or errors occur. This flag may be specified multiple times with different parameters that combine meaningfully. If *param* is a positive number, it specifies the maximum backtrace depth, abbreviating deeper ones. Other valid parameter values are the following:

  `0` Do not limit the maximum backtrace depth; this is the default.

  `all` Force all locations to be printed, even "quiet" ones (see "Excluding locations from backtraces" in *gbasm*(5) for details).

  `no-all`
  : Do not print "quieted" locations in backtraces; this is the default.

  `collapse`
  : Print all locations on one line.

  `no-collapse`
  : Print one location per line; this is the default.

`-b` *chars*, `--binary-digits` *chars*
: Allow two characters to be used for binary constants in addition to the default '0' and '1'. Valid characters are numbers other than '0' and '1', letters, '.', '#', or '@'.

`--color` *when*
: Specify when to highlight warning and error messages with color: `always`, `never`, or `auto`. `auto` determines whether to use colors based on the *NO_COLOR*: https://no-color.org/ or *FORCE_COLOR*: https://force-color.org/ environment variables, or whether the output is to a TTY.

`-D` *name*[=*value*], `--define` *name*[=*value*]
: Add a string symbol to the compiled source code. This is equivalent to *name* **EQUS** "*value*" in code, or *name* **EQUS** "1" if *value* is not specified.

`-E`, `--export-all`
: Export all labels, including unreferenced and local labels.

`-g` *chars*, `--gfx-chars` *chars*
: Allow four characters to be used for graphics constants in addition to the default '0', '1', '2', and '3'. Valid characters are numbers other than '0' to '3', letters, '.', '#', or '@'. The defaults are 0123.

`-h`, `--help`
: Print help text for the program and exit. (Help text wraps to the value of the `COLUMNS` environment variable if that is defined as nonzero; or else to the console window width if output is to a TTY.)

`-I` *path*, `--include` *path*
: Add a new "include path"; *path* must point to a directory. When any **INCLUDE** (including the implicit one from `-P`), **INCBIN**, or **READFILE** is attempted, **rgbasm** first looks up the provided path from its working directory; if this fails, it tries again from each of the "include path" directories, in the order they were provided.

`-M` *depend_file*, `--dependfile` *depend_file*
: Write *make*(1) dependencies to *depend_file*.

`-MG`
: To be used in conjunction with `-M`. This makes **rgbasm** assume that missing files are auto-generated: when any **INCLUDE** (including the implicit one from `-P`), **INCBIN**, or **READFILE** is attempted on a non-existent file, it is added as a dependency, then **rgbasm** exits normally or continues processing (depending on whether `-MC` was enabled) instead of erroring out. This feature is used in automatic updating of Makefiles.

`-MC`
: Implies `-MG`. This makes **rgbasm** continue processing after a non-existent dependency file, instead of exiting. Note that this is*not* recommended if any non-existent dependencies would have influenced subsequent processing, e.g. by causing an **IF** condition to take a different branch.

`-MP`
: When enabled, this adds a phony target to the rules emitted by `-M` for each dependency other than the main file. This prevents *make*(1) from erroring out when dependency files are deleted.

`-MT` *target_file*
: Add a target to the rules emitted by `-M`. The exact string provided will be written, including spaces and special characters.

  `-MT fileA -MT fileB`

  is equivalent to

  `-MT 'fileA fileB'`.

  If neither this nor `-MQ` is specified, the output file name is used.

`-MQ` *target_file*
: Same as `-MT`, but additionally escapes any special *make*(1) characters, essentially '$'.

`-o` *out_file*, `--output` *out_file*
: Write an object file to the given filename.

`-P` *include_file*, `--preinclude` *include_file*
: Pre-include a file. This acts as if a **INCLUDE** "*include_file*" was read before the input *asmfile*. Multiple files can be pre-included in the order they were provided.

`-p` *pad_value*, `--pad-value` *pad_value*
: Use this as the value for **DS** directives in ROM sections, unless overridden. The default is 0x00.

`-Q` *fix_precision*, `--q-precision` *fix_precision*
: Use this as the precision of fixed-point numbers after the decimal point, unless they specify their own precision. The default is 16, so fixed-point numbers are Q16.16 (since they are 32-bit integers). The argument may start with a '`.`' to match the Q notation, for example, `-Q` *.16*.

`-r` *recursion_depth*, `--recursion-depth` *recursion_depth*
: Specifies the recursion depth past which **rgbasm** will assume being in an infinite loop. The default is 64.

`-s` *features*:*state_file*, `--state` *features*:*state_file*
: Write the specified *features* to *state_file*, based on the final state of **rgbasm** at the end of its input. The expected *features* are a comma-separated subset of the following:

  `equ` Write all numeric constants as **def** *name* **equ** *value*.

  `var` Write all variables as **def** *name* **=** *value*.

  `equs` Write all string constants as **def** *name* **equs** "*value*".

  `char` Write all characters as **charmap** *name*, *value*.

  `macro` Write all macros as **macro** *name* ... **endm**.

  `all` Acts like `equ,var,equs,char,macro`.

  This flag may be specified multiple times with different feature subsets to write them to different files (see "EXAMPLES" below).

`-V`, `--version`
: Print the version of the program and exit.

`-v`, `--verbose`
: Be verbose. The verbosity level is increased by one each time the flag is specified, with each level including the previous:
  1. Print the **rgbasm** configuration before taking actions.
  2. Print a notice before significant actions.
  3. Print some of the actions' intermediate results.
  4. Print some internal debug information.
  5. Print detailed internal information.

  The verbosity level does not go past 6.

  Note that verbose output is only intended to be consumed by humans, and may change without notice between RGBDS releases; relying on those for scripts is not advised.

`-W` *warning*, `--warning` *warning*
: Set warning flag *warning*. A warning message will be printed if *warning* is an unknown warning flag. See the "DIAGNOSTICS" section for a list of warnings.

`-w`
: Disable all warning output, even when turned into errors.

`-X` *max_errors*, `--max-errors` *max_errors*
: If more than this number of errors (not warnings) occur, then abort the assembly process; `-X` *0* disables this behavior. The default is 100 if **rgbasm** is printing errors to a terminal, and 0 otherwise.

@at_file
: Read more options and arguments from a file, as if its contents were given on the command line. Arguments are separated by whitespace or newlines. Lines starting with a hash sign ('#') are considered comments and ignored.

  No shell processing is performed, such as wildcard or variable expansion. There is no support for escaping or quoting whitespace to be included in arguments. The standard '`--`' to stop option processing also disables at-file processing. Note that while '`--`' can be used*inside* an at-file, it only

disables option processing within that at-file, and processing continues in the parent scope.

DIAGNOSTICS

Warnings are diagnostic messages that indicate possibly erroneous behavior that does not necessarily compromise the assembling process. The following options alter the way warnings are processed.

`-Werror`
: Make all warnings into errors. This can be negated as `-Wno-error` to prevent turning all warnings into errors.

`-Werror=`
: Make the specified warning or meta warning into an error. A warning's name is appended (example: `-Werror=obsolete`), and this warning is implicitly enabled and turned into an error. This can be negated as `-Wno-error=` to prevent turning a specified warning into an error, even if `-Werror` is in effect.

The following warnings are "meta" warnings, that enable a collection of other warnings. If a specific warning is toggled via a meta flag and a specific one, the more specific one takes priority. The position on the command-line acts as a tie breaker, the last one taking effect.

`-Wall`
: This enables warnings that are likely to indicate an error or undesired behavior, and that can easily be fixed.

`-Wextra`
: This enables extra warnings that are less likely to pose a problem, but that may still be wanted.

`-Weverything`
: Enables literally every warning.

The following warnings are actual warning flags; with each description, the corresponding warning flag is included. Note that each of these flags also has a negation (for example, `-Wobsolete` enables the warning that `-Wno-obsolete` disables; and `-Wall` enables every warning that `-Wno-all` disables). The flags are listed alphabetically below.

`-Wassert`
: Warn when **WARN**-type assertions fail. (See "Aborting the assembly process" in *rgbasm*(5) for **ASSERT**). This warning is enabled by default.

`-Wbackwards-for`
: Warn when **FOR** loops have their start and stop values switched according to the step value. This warning is enabled by `-Wall`.

`-Wbuiltin-args`
: Warn about incorrect arguments to built-in functions, such as **STRSLICE**() with indexes outside of the string's bounds. This warning is enabled by `-Wall`.

`-Wcharmap-redef`
: Warn when re-defining a charmap mapping. This warning is enabled by `-Wall`.

`-Wdiv`
: Warn when dividing the smallest negative integer (-2**31) by -1, which yields itself due to integer overflow.

`-Wempty-data-directive`
: Warn when **DB**, **DW**, or **DL** is used without an argument in a ROM section. This warning is enabled by `-Wall`.

`-Wempty-macro-arg`
: Warn when a macro argument is empty. This warning is enabled by `-Wextra`.

`-Wempty-strrpl`
: Warn when **STRRPL**() is called with an empty string as its second argument (the substring to replace). This warning is enabled by `-Wall`.

`-Wexport-undefined`
: Warn when exporting an undefined symbol. This warning is enabled by `-Wall`.

`-Wlarge-constant`
: Warn when a constant too large to fit in a signed 32-bit integer is encountered. This warning is enabled by default.

`-Wmacro-shift`
: Warn when shifting macro arguments past their limits. This warning is enabled by `-Wextra`.

`-Wnested-comment`
: Warn when the block comment start sequence '`/*`' is found inside of a block comment. Block comments cannot be nested, so the first '`*/`' will end the whole comment. This warning is enabled by default.

`-Wobsolete`
: Warn when obsolete features are encountered, which have been deprecated and may later be removed. This warning is enabled by default.

`-Wnumeric-string=`
: Warn when a multi-character string is treated as a number. `-Wnumeric-string=0` or `-Wno-numeric-string` disables this warning. `-Wnumeric-string=1` or just `-Wnumeric-string` warns about strings longer than four characters, since four or fewer characters fit within a 32-bit integer. `-Wnumeric-string=2` warns about any multi-character string.

`-Wpurge=`
: Warn when purging symbols which are likely to have been necessary. `-Wpurge=0` or `-Wno-purge` disables this warning. `-Wpurge=1` warns when purging any exported symbol (regardless of type). This warning level is enabled by default. `-Wpurge=2` or just `-Wpurge` also warns when purging any label (even if not exported). This warning level is enabled by `-Wall`.

`-Wshift`
: Warn when shifting right a negative value. Use a division by 2**N instead.

`-Wshift-amount`
: Warn when a shift's operand is negative or greater than 32.

`-Wtruncation=`
: Warn when an implicit truncation (for example, **db** to an 8-bit value) loses some bits. `-Wtruncation=0` or `-Wno-truncation` disables this warning. `-Wtruncation=1` or just `-Wtruncation` warns when an N-bit value is 2**N or greater, or less than -2**N. This warning level is enabled by default. `-Wtruncation=2` also warns when an N-bit value is less than -2**(N-1), which will not fit in two's complement encoding. This warning level is enabled by `-Wextra`.

`-Wunmapped-char=`
: Warn when a character goes through charmap conversion but has no defined mapping. `-Wunmapped-char=0` or `-Wno-unmapped-char` disables this warning. `-Wunmapped-char=1` or just `-Wunmapped-char` only warns if the active charmap is not empty. This warning level is enabled by default. `-Wunmapped-char=2` warns if the active charmap is empty, and/or is not the default charmap 'main'. This warning level is enabled by `-Wall`.

`-Wunmatched-directive`
: Warn when a **`PUSHC`**, **`PUSHO`**, or **`PUSHS`** directive does not have a corresponding **`POPC`**, **`POPO`**, or **`POPS`**. This warning is enabled by `-Wextra`.

`-Wunterminated-load`
: Warn when a **`LOAD`** block is not terminated by an **`ENDL`**. This warning is enabled by `-Wextra`.

`-Wuser`
: Warn when the **`WARN`** built-in is executed. (See “Aborting the assembly process” in *rgbasm*(5) for **`WARN`**). This warning is enabled by default.

EXAMPLES

You can assemble a source file in two ways.

Straightforward way:

```
$ rgbasm -o bar.o foo.asm
```

Pipes way:

```
$ cat foo.asm | rgbasm -o bar.o -
$ rgbasm -o bar.o - < foo.asm
```

The resulting object file is not yet a usable ROM image—it must first be run through *rgblink*(1) and then *rgbfix*(1).

Writing the final assembler state to a file:

```
$ rgbasm -s all:state.dump.asm foo.asm
```

Or to multiple files:

```
$ rgbasm -s equ,var:numbers.dump.asm -s equs:strings.dump.asm foo.asm
```

BUGS

Please report bugs or mistakes in this documentation on *GitHub*: https://github.com/gbdev/rgbds/issues.

SEE ALSO

rgbasm(5), *rgblink*(1), *rgbfix*(1), *rgbgfx*(1), *gbz80*(7), *rgbasm-old*(5), *rgbds*(5), *rgbds*(7)

HISTORY

`rgbasm` was originally written by Carsten Sørensen as part of the ASMotor package, and was later repackaged in RGBDS by Justin Lloyd. It is now maintained by a number of contributors at https://github.com/gbdev/rgbds.